

Acute Effects of Combined Lower-Body High-Intensity Interval Training and Upper-Body Strength Exercise on Explosive Strength Performance in Naturally Menstruating Women

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Purpose: We aimed at investigating the acute effects of lower-body high-intensity interval training (HIIT) on upper- and lower-body explosive strength assessed by mean propulsive velocity (MPV) in naturally menstruating women. In addition, we assessed the combination of lower-body HIIT and squat, as well as lower-body HIIT and bench press, on bench press and squat MPV. **Methods:** Thirteen women (age: 23 [2] y, menstrual cycle length: 28.4 [2.0] d) completed 2 training modalities on separate days (separated by 30 [4.2] d) consisting of HIIT followed by lower-body (HIIT + LBS) or upper-body (HIIT + UBS) strength loading. Squat and bench press MPV were assessed before HIIT (T0), after HIIT (T1), after the strength loading (T2), and 24 hours postloading (T3). **Results:** Mixed factorial analysis of variance indicated a significant effect for time in bench press and squat ($P < .001$) but not for interaction. Pairwise comparison showed that bench press MPV remained unchanged ($P = 1.000$) at T1 but was reduced at T2 compared with T0 (HIIT + LBS: -8.2% [3.9%], HIIT + UBS: -13.8% [12.1%], $P < .001$) and T1 (HIIT + LBS: -7.1% [3.2%], HIIT + UBS: -12.7% [8.7%], $P < .001$). Squat MPV decreased at T1 (HIIT + LBS: -6.0% [8.8%], HIIT + UBS: -4.8% [5.4%], $P = .009$) and was found to be decreased at T2 compared with T0 in both conditions (HIIT + LBS: -6.9% [3.3%], HIIT + UBS: -7.4% [6.1%], $P < .001$) but not compared with T1 ($P = 1.000$). Bench press and squat MPV returned to baseline at T3 compared with T0 ($P > .050$). **Conclusion:** Lower- but not upper-body explosive strength was reduced by HIIT. HIIT combined with upper- or lower-body strength loading resulted in a reduction of squat and bench press explosive strength.

Keywords: concurrent training, menstrual cycle, strength training

During the past decade, several studies have shown that concurrent endurance and strength training does not lead to impaired adaptations in maximal strength and endurance capacity compared to strength or endurance training alone.^{1,2} However, in a recent meta-analysis, we showed that the development of explosive strength is impaired, especially when both endurance and strength training are combined within the same session.³ However, this interference in explosive strength development has been demonstrated exclusively in studies examining lower-body training.^{4,5} Since some sports also require a combination of upper-body strength and lower-body endurance,⁶ it is important to clarify whether fatigue induced by endurance exercises of the lower-body acutely impairs strength performance of the upper body.

Indeed, a few studies have examined the acute effects of lower-body endurance exercise on selected measures of strength performance in the upper body (eg, maximal strength and strength endurance)^{7–10} but, there is a lack of evidence on the effects of lower-body endurance exercise on upper-body explosive strength performance. Considering that explosive strength is a key performance parameter in numerous sports and is particularly susceptible to neuromuscular fatigue,^{11,12} assessing possible interference of lower-body endurance and upper-body explosive strength is

indispensable. We have previously shown that in men lower-body high-intensity interval training (HIIT) does not acutely decrease upper-body explosive strength, whereas lower-body HIIT in combination with both upper- or lower-body strength exercise did lead to reductions in explosive bench press performance.¹³ Thus, it appears that adding lower-body HIIT to strength exercise compromises subsequent explosive strength performance regardless of the muscle group though it remains unclear whether this is the case in women as well.

Sex-based differences in fatigability have been investigated in the past, showing that women and men display profound differences in neuromuscular performance and fatigability, with the consensus being that women tend to be less fatigable than men.^{14,15} The mechanisms underlying differences in fatigability are manifold and relate to differences in physiology and anatomy, such as muscle mass, blood flow and muscle perfusion, muscle fiber-type distribution, and variations in reproductive hormones.¹⁵ Especially due to differences in the hormonal profile, a transfer of data obtained in men to women is not without difficulties.¹⁶ Particularly in naturally menstruating women, concentrations of the female sex hormones estrogen and progesterone fluctuate throughout the menstrual cycle. In the early follicular phase, both estrogen and progesterone levels are low, followed by the late follicular phase characterized by high estrogen and low progesterone levels, and the luteal phase with both high levels of estrogen and progesterone.¹⁷ In a recent study, it was shown that eumenorrheic women showed differences in central drive across the menstrual cycle phases following a power-type loading (ie,

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2 sets of 10 repetitions at 60% 1-repetition maximum [1RM] with 2-min rest between sets).¹⁸ Thus, it seems that fluctuations in the ovarian hormones may alter the capacity to generate explosive strength, however, it remains unknown whether this will also affect possible acute interference of endurance and strength exercises.

To investigate the acute interactions of lower- and upper-body endurance and strength exercises in women, this study had 2 objectives: (1) to examine the acute effects of lower-body HIIT on lower-body (ie, squat) and upper-body (ie, bench press) explosive strength, and (2) to assess the acute effects of combined HIIT and bench press versus squat loading on both bench press and squat explosive strength performance. To minimize the potential effects of ovarian hormones, the experimental procedures were conducted in the follicular phase. Based on previously obtained data, we hypothesized that HIIT induces an acute reduction in squat, but not in bench press explosive strength. Furthermore, we hypothesized that acute concurrent HIIT and squat loading, as well as HIIT and bench press loading would lead to reductions in both squat and bench press explosive strength. Since we previously examined the effects using the same study design,¹³ we further hypothesize that the magnitude of reductions in explosive strength in women might be lower to that previously shown in men.

Materials and Methods

Participants

The sample size was defined a priori based on the main outcome, that is, reductions in mean propulsive velocity (MPV) following the endurance and combined loadings. The projected sample size was assessed by G*Power 3.1,¹⁹ with $\alpha = .05$, an effect size of 0.5 ($\eta_p^2 = .220$) as determined previously,¹³ and a power of 0.95. The analysis revealed a required minimum of 12 subjects. Hence, 13 physically active women (age: 23 [2] y; height: 167.3 [6.5] cm; body mass: 62.8 [6.0] kg; body mass index: 22.4 [1.5] kg·m⁻²; menstrual cycle length: 28.4 [2.0] across 3 menstrual cycles; relative squat 1RM 1.1 [0.2] kg·kg⁻¹ body weight; relative bench press 1RM 0.7 [0.1] kg·kg⁻¹ body weight; maximal aerobic power [MAP] 3.7 [0.3] W·kg⁻¹ body weight) volunteered to participate in

this study. Participants completed a health questionnaire and a familiarization session where they were informed about the study design and the inclusion criteria. Women were eligible to participate in the study if they were (1) nonsmokers, (2) without chronic or acute injuries, (3) between the age of 18 and 40 years, (4) physically active, had at least 2 years of experience in strength training and were not specifically endurance trained, (5) had a body mass index between 18 and 25 kg·m⁻², and (6) reported regular menstrual bleedings at least in 3 consecutive months prior to participation. Information regarding the menstrual cycle and monthly bleeding was obtained during the familiarization interview. Participants were informed about possible risks associated with the study and provided written informed consent prior to inclusion. This study was approved by the local institutional board (036/2019) and was conducted according to the Declaration of Helsinki.

Study Design

This study design has been adapted from a previous study that we conducted in men.¹³ All participants participated in 2 experimental sessions in a randomized order consisting of HIIT + lower-body strength loading (HIIT + LBS) and HIIT + upper-body strength loading (HIIT + UBS). In order to quantify the individual endurance and strength exercise load, an initial graded exercise test on a stationary bike and 1RM test in the squat and bench press were performed. Participants visited the laboratory at least 1 month prior to the start of the testing period for a familiarization session. Thereafter, initial endurance testing commenced within the late luteal phase of the menstrual cycle, while the 1RM test was performed within the subsequent menstrual cycle (2–5 d). Since it was previously reported that in naturally menstruating women differences in maximal voluntary force and rate of force development might be dependent on the cycle phase,¹⁸ the experimental sessions (acute test) were conducted exclusively in the follicular phase within 2 subsequent menstrual cycles (Figure 1; 5–11 d) in order to minimize the influence of ovarian hormones on the explosive strength. The timing of the measurement (eg, 1RM test and acute test) was determined based on the days after the onset of menstruation. Therefore, each participant completed the

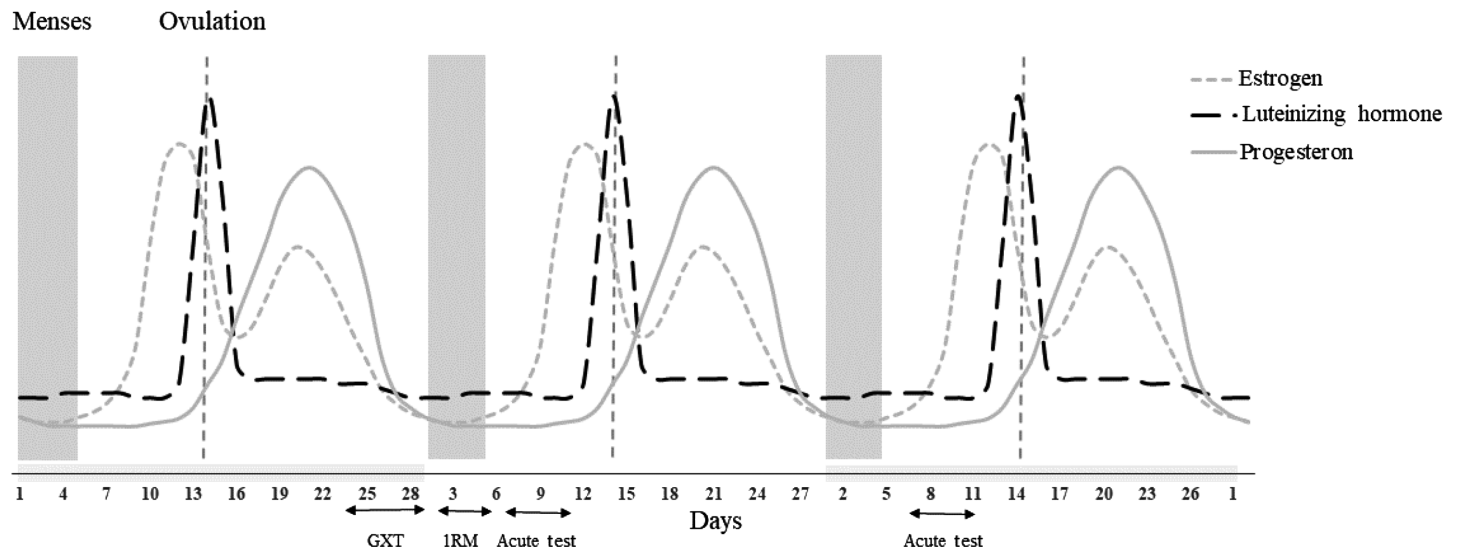


Figure 1 — Timing of the corresponding strength (1RM) and acute testing within a 28-day natural menstrual cycle. Adapted from Elliott-Sale et al.²⁰ 1RM indicates incremental 1-repetition maximum test; GXT, graded exercise test.

acute test on the same days of the respective cycle. Sessions were separated by 30 (4.2) days.

During this period, participants were instructed to maintain their regular exercise program, but to refrain from intense exercise for 24 hours prior to each session. The experimental sessions were conducted in the morning, and time of day was kept the same for each participant (± 1 h).

Measurements

The following measurements (ie, endurance testing, strength testing, explosive strength assessment, and experimental sessions) have been described in detail elsewhere.¹³ A summary of the tests performed is provided below.

Endurance Testing. To assess the basal MAP, participants completed a graded exercise test on an electronically braked cycle ergometer (Lode Excalibur Sport) until volitional exhaustion.

Strength Testing. To assess basal strength performance, participants performed an incremental (1RM) test in both squat and bench press, respectively. Along with the MPV, the eccentric displacement was obtained during each repetition of the test using a linear velocity transducer (T-Force System, Ergotech Consulting, S.L).

Explosive Strength Assessment. To determine the acute effects of endurance and strength exercises, explosive strength performance was assessed in bench press and squats using the MPV. The test commenced within 30 seconds following the endurance and strength loading, respectively. Participants were requested to perform 3 repetitions at a load of 60% 1RM as fast as possible. In order to collect baseline values, before, and 24 hours after the experimental session, 3 $\times$ 3 repetitions of both squat and bench press, separated by 2 minutes of interset rest were performed.

Experimental Session Procedure. The participants arrived at the laboratory after an overnight fast and received a maltodextrin solution 30 minutes before the experimental loading (50 g maltodextrin dissolved in 250 mL water). Thereafter, participants were required to complete a 5-minute warm-up on a cycle ergometer. Initially, long high-intensity intervals were performed on a cycle ergometer (4 $\times$ 4 min at 80% MAP). After 5 minutes of recovery, a strength loading consisting of 3 $\times$ 5 followed by 3 $\times$ 3 repetitions of either squat (HIIT + LBS) or bench press exercise (HIIT + UBS) was performed. The load was set at 80% of the 1RM, with 2 minutes of rest between the sets. Participants were requested to execute a maximal movement velocity during the concentric phase. Explosive strength was assessed before HIIT (T0) and immediately after HIIT (T1) as well as immediately after the strength loading (T2) and 24 hours after (T3). In addition, movement velocity was recorded throughout the strength loading (T-Force Dynamic Measurement System, version 2.3 Ergotech Consulting, S.L) and velocity loss (Δ MPV) was calculated for each set between the fastest and slowest repetition of each set.

Statistical Analysis

Normality of distribution was assessed by the Shapiro–Wilk test. A mixed factorial analysis of variance with repeated measures (time: T0, T1, T2, T3; interaction: HIIT + LBS, vs HIIT + UBS) was used to analyze differences in the MPV during explosive strength

assessment. In case the underlying assumption of sphericity was not fulfilled, Greenhouse–Geisser correction was applied. Post hoc testing was performed using Bonferroni correction. Effect sizes for main effects were expressed as partial eta squared (η_p^2). The statistical analysis was performed using SPSS Statistics (version 27, IBM). The level of significance for all tests was set at $P \leq .050$. Data are presented as mean (SD).

Results

Mixed factorial analysis of variance indicated a significant effect for time (squat: $F_{1,887,45.298} = 10.714$, $P < .001$, $\eta_p^2 = .309$; bench press: $F_{3,72} = 20.921$, $P < .001$, $\eta_p^2 = .466$); but interaction effects were neither found for squat MPV ($F_{1,887,45.298} = 0.489$, $P = .606$, $\eta_p^2 = .020$) nor bench press MPV ($F_{3,72} = 1.592$, $P = .199$, $\eta_p^2 = .062$; Figure 2). Pairwise comparison showed that squat MPV statistically decreased at T1 following HIIT (HIIT + LBS: -6.0% [8.8%], HIIT + UBS: -4.8% [5.4%], $P = .009$), while bench press MPV remained statistically unchanged (HIIT + LBS: -1.1% [4.6%], HIIT + UBS: -1.2% [10.0%], $P = 1.000$). Squat MPV was found to be decreased at T2 compared with T0 in both conditions (HIIT + LBS: -6.9% [3.3%], HIIT + UBS: -7.4% [6.1%], $P < .001$) but not compared with T1 (HIIT + LBS: -0.09% [10.7%], HIIT + UBS: -2.4% [9.0%], $P = 1.000$). Bench press MPV was statistically reduced at T2 after both loadings compared with T0 (HIIT + LBS: -8.2% [3.9%], HIIT + UBS: -13.8% [12.1%]; $P < .001$) and T1 (HIIT + LBS: -7.1% [3.2%], HIIT + UBS: -12.7% [8.7%], $P < .001$). However, MPVs returned to baseline at T3 compared with T0 in both bench press ($P = .261$) and squat ($P = 1.000$).

In both conditions, a significant difference was found between the absolute MPV of the first and last repetition of each set ($F_{2,44} = 46.866$, $P < .001$; Figure 3). Δ MPV was significantly greater in the third set (ie, performed with 5 repetitions) compared to sets 4 through 6 (ie, performed with 3 repetitions; $P < .037$). However, Δ MPV across all sets was not different between squat (Δ MPV: -19.6% [11.0%]) and bench press (Δ MPV: -24.6% [11.3%]; $F_{5,110} = 1.031$, $P = .392$).

Discussion

Our data indicate that in naturally menstruating women, HIIT alone did not acutely affect upper- but led to reductions in lower-body explosive strength performance. However, both concurrent HIIT and bench press, as well as concurrent HIIT and squat loading resulted in an acute decrease of both explosive bench press and squat performance that returned to baseline within 24 hours. The magnitude of reductions observed in bench press and squat performance following the combined loadings was similar to that previously observed by us in men.

In the present study, HIIT did not result in an acute decrease in explosive bench press performance but did lead to reductions in explosive squat performance. This result is consistent with what we previously showed in men, where explosive strength in the squat was also acutely decreased after HIIT, while bench press explosive strength remained unchanged.¹³ However, the finding of reduced lower-body explosive strength following HIIT differs somewhat from previous findings in women, where no reductions in the rate of force development in isometric leg press following 60 minutes of running at a moderate intensity were shown, while in men explosive strength was statistically reduced.¹⁴ It is likely that this

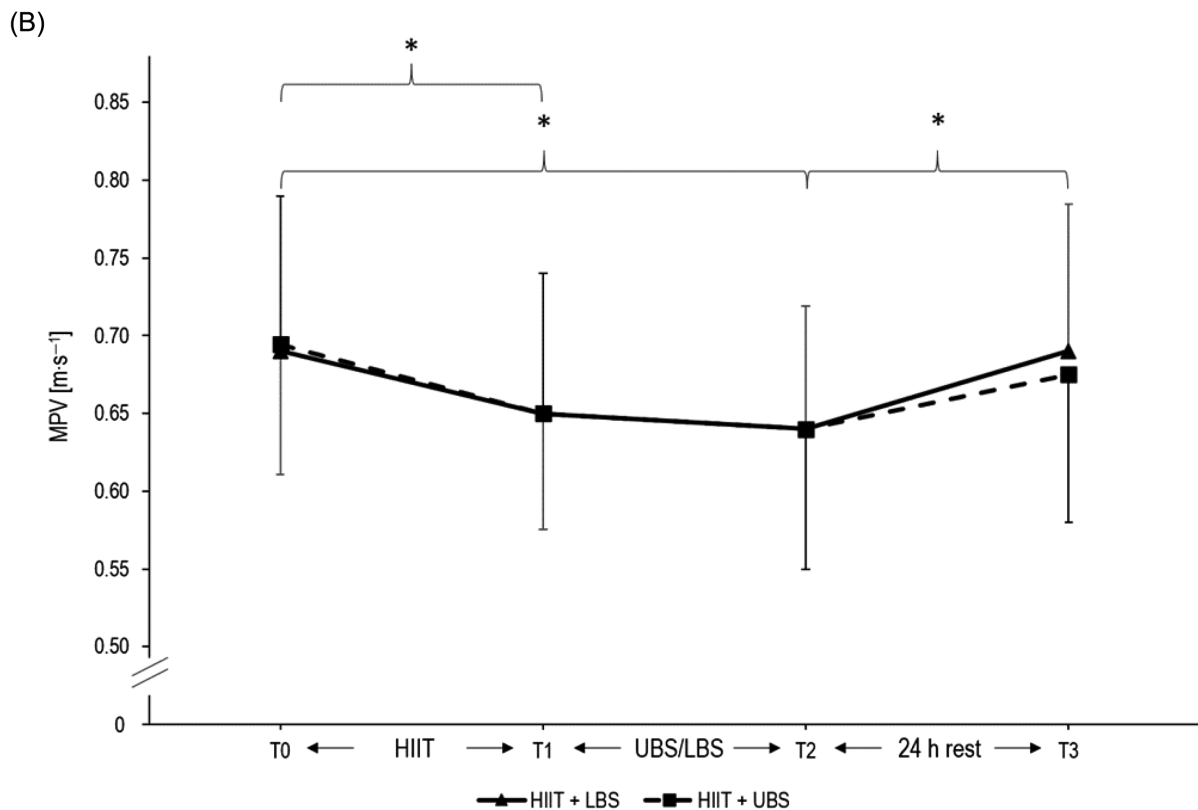
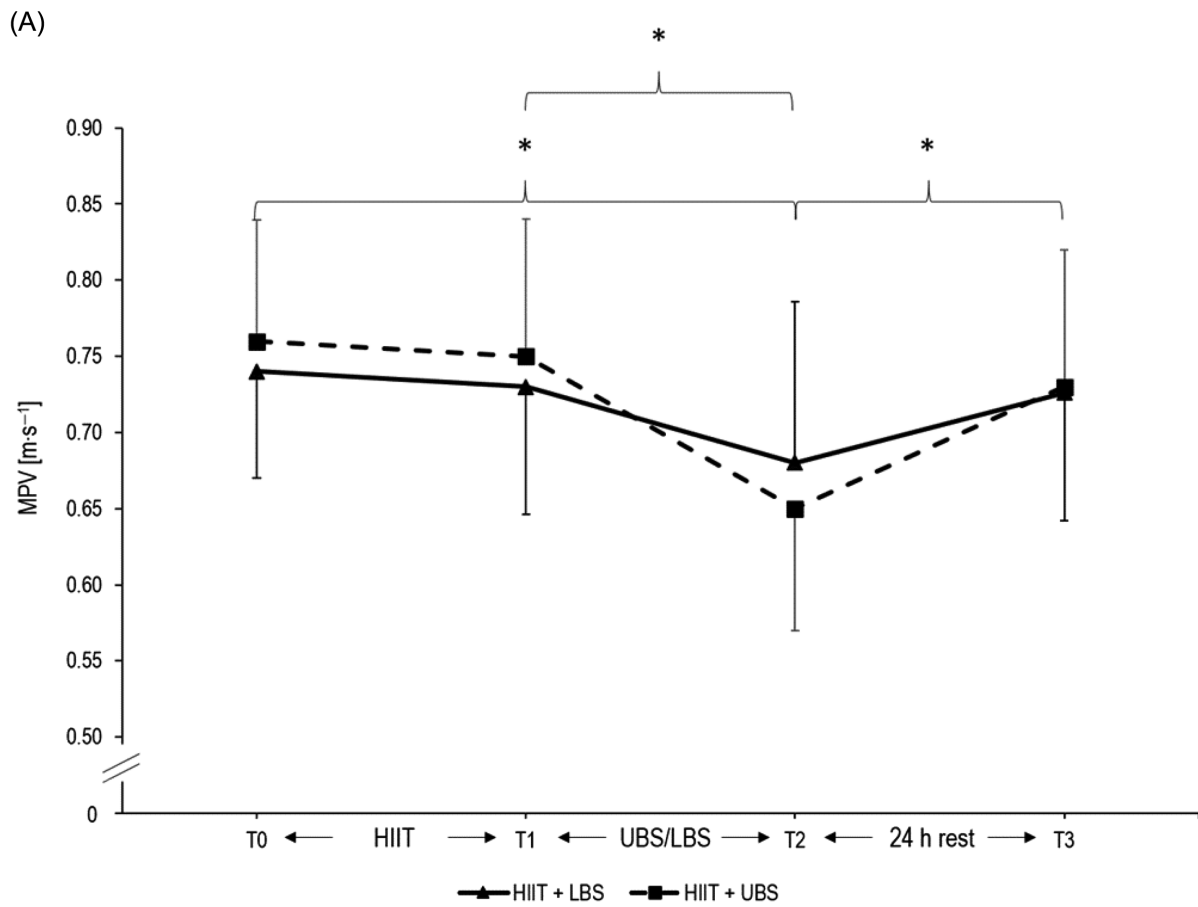


Figure 2 — Absolute values of bench press (A) and squat MPV (B) during the HIIT + LBS session (solid line) and the HIIT + UBS session (dashed line). HIIT indicates high-intensity interval endurance loading; HIIT + LBS, lower-body high-intensity interval cycling + squat loading; HIIT + UBS, upper-body high-intensity interval cycling + bench press loading; LBS, lower-body strength loading; MPV, mean propulsive velocity; UBS, upper-body strength loading. *Significant difference ($P \leq .050$) in absolute MPV during both sessions.

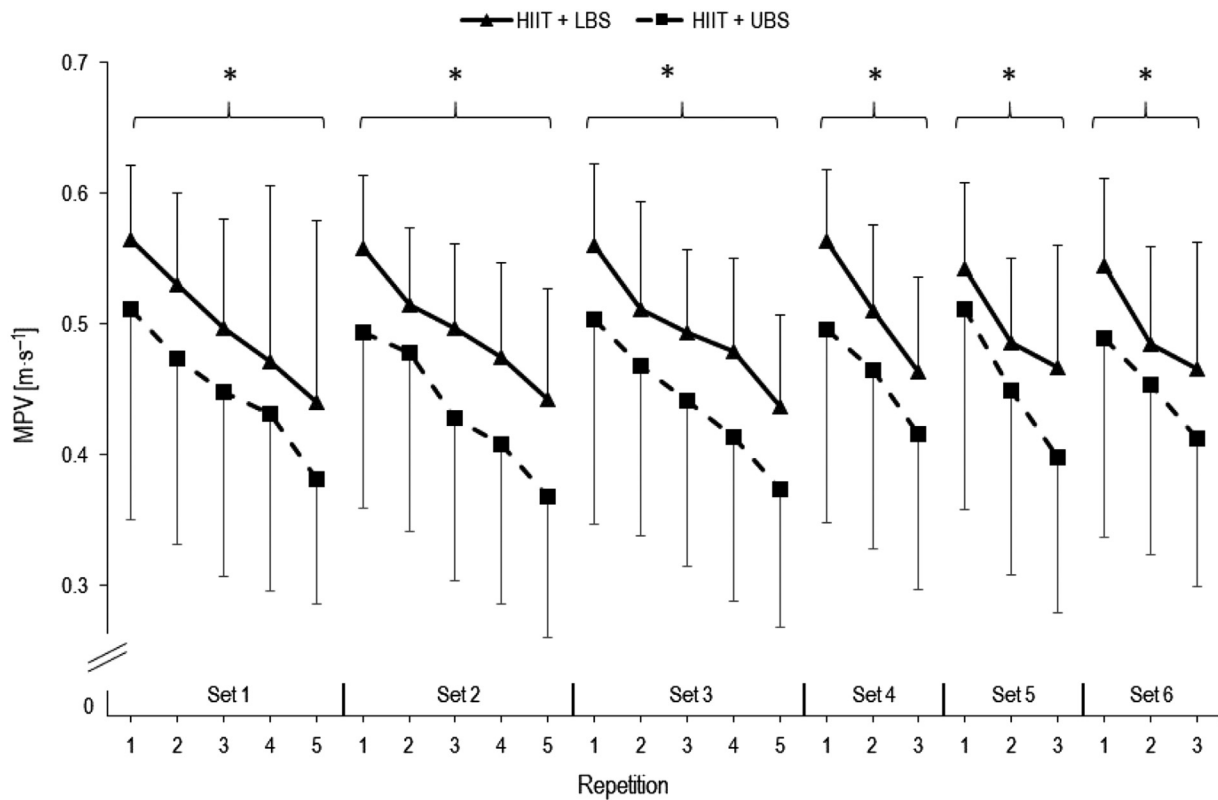


Figure 3 — Absolute MPV values of all repetitions during squat loading in the HIIT + LBS (solid line) and bench press loading during the HIIT + UBS session (dotted line). *Significant difference ($P \leq .050$) in absolute MPV during both sessions. HIIT indicates high-intensity interval endurance loading; LBS, lower-body strength loading; MPV, mean propulsive velocity; UBS, upper-body strength loading.

discrepancy may at least partially be explained by the endurance protocols (ie, running vs cycling). The endurance protocol in the present study was specific to the muscle group of the explosive strength test, whereas running has been shown to induce less loads on thighs compared to cycling.²¹ Similarly, in the present study the explosive strength was tested dynamically whereas, in the aforementioned study the explosive strength was evaluated using isometric rate of force development at one specific angle only. Importantly, it should be noted that we used long high-intensity intervals on a bicycle ergometer in the present study and that the influence on explosive strength may differ with the endurance training modality (eg, cycling vs running).²² It also remains questionable whether the observed acute effects lead to a chronic maladaptation of explosive strength.

Interestingly, the magnitude of fatigue in women following HIIT was comparable with that previously observed by us for squat in men (-5.2% [7.6%] vs -7.1% [4.8%]).¹³ This finding is, to some extent, surprising, since it has been reported that women tend to be less fatigable than men.²³ However, the differences in fatigability between men and women are also thought to depend on the exercise intensity, whereby differences in fatigability are more pronounced at lower than at near-maximal intensities.²⁴ Differences in fatigability at lower intensities between men and women may be caused by distinct metabolic and mechanical attributes. For example, men dispose greater muscle mass, resulting in poorer blood flow and attenuated muscle perfusion compared with women.²⁵ In addition, women have a greater proportion of type I fibers, which are more fatigue resistant than type II fibers.²⁶ Together these factors show that sex-based

differences appear to be especially pronounced in submaximal contractions. Since we used heavy loads for the strength loading (ie, $3 \times 5 + 3 \times 3$ at 80% of 1RM), the marginal differences in fatigue levels could be explained by the high exercise intensities.

Similar to our previous findings in men, both combined loadings (ie, HIIT + LBS and HIIT + UBS) led to reductions in explosive strength performance in bench press and squat when compared to T0 but returned to baseline within 24 hours. While MPV during the bench press was affected by both lower- and upper-body strength loadings, the magnitude of reductions in MPV was somewhat larger for HIIT + UBS (-13.8% [12.1%]) compared to HIIT + LBS (-8.2% [3.9%]), though not statistically different. Furthermore, the magnitude of reductions in bench press and squat performance was also similar to that previously observed in men (ie, men: HIIT + UBS -14.4% [7.4%]; HIIT + LBS -6.8% [4.6%], respectively). In contrast to explosive bench press performance, MPVs during the squat assessment were not further reduced by both strength loadings. Thus, squat explosive strength was not significantly reduced beyond the fatigue induced by HIIT. This finding is similar to our data obtained in men, showing that when a plateau of fatigue is induced a strength loading might not generate additional fatigue.¹³

The mechanisms leading to the reduction in explosive strength, however, are unknown. Exercise-induced muscle fatigue may have several causes. It can result not only from peripheral changes at the muscular level (eg, ionic metabolite accumulation that inhibits cross-bridge formation and activation)²⁷ but also from the central nervous system, failing to drive the motoneurons sufficiently via inhibited spinal and supraspinal factors (central

fatigue).²⁸ Considering that in the present study HIIT did lead to reductions in explosive squat but not explosive bench press performance, it appears that peripheral rather than central fatigue might be the cause for the observed reductions. Conversely, our data indicate that a combination of lower-body HIIT and strength performance possibly leads to peripheral and central fatigue, causing reductions in explosive strength even in the unloaded muscle group. Since it was previously shown that in men and women a strength loading induces peripheral and central fatigue,²⁹ the observed reductions in bench press explosive strength are most likely induced by fatigue resulting from the strength loading. However, it cannot be ruled out that the preceding HIIT also led to delayed fatigue via metabolic disturbances, which, in turn, could have affected the upper-body explosive strength, or that the overall session length additionally contributed to the reduction in bench press explosive strength following the combined loading.

In addition, this study assessed the Δ MPV in both strength loadings. Intriguingly, we could not find greater Δ MPV in the bench press loading compared to the squat loading (bench press Δ MPV: -24.6% [11.3%], squat Δ MPV: -19.6% [11.0%], $P = .392$). Since it is well known that in men Δ MPV in the upper extremities is greater compared to the lower extremities¹¹ when using the same relative intensity (ie, % 1RM), one could assume that this might be the case in women as well. While in men greater Δ MPV has been attributed to greater proportion of type II fibers in the upper body compared to the lower body, women in general show larger proportions of fatigue-resistant muscle fibers and tend to have a lower proportion of their muscle mass distributed in the upper body,²⁶ which might explain the similar Δ MPV in the upper- and lower-body observed in the present study.

Although the present study expands on previous evidence on the acute effects of combined endurance and strength training, some limitations need to be acknowledged. The experimental sessions were conducted within the follicular phase of the menstrual cycle, and previous studies have shown that fluctuations in ovarian hormones throughout the menstrual cycle affect the neural excitability³⁰ as well as voluntary activation.³¹ Whether this affects neuromuscular performance remains a matter of debate. For example, Peltonen et al¹⁸ recently showed that fatigue in eumenorrheic women, particularly central fatigue, is associated with estrogen and progesterone levels. While central fatigue seems to be larger in participants with lower progesterone levels, increased estradiol levels have been associated with greater neural excitation.³⁰ However, despite changes in central fatigue, Peltonen et al¹⁸ could not find differences in power production or maximal strength between the menstrual cycle phases. Nevertheless, considering that the experimental sessions in the present study were conducted in the follicular phase (5–11 d) of the menstrual cycle (ie, high/rising estrogen and low progesterone levels), our findings may not necessarily be generalized. Furthermore, in the present study we did not report ovarian hormone concentrations, and thus, we are unable to relate hormone concentrations with the individual magnitude in reductions of explosive strength. Since intraindividual hormone concentrations vary both throughout the cycle as well as within each cycle phase,³² individual data on the hormone concentrations may improve the understanding of the relationship between strength capacity and hormone concentrations. However, since the timing of the acute test was standardized for each participant, the hormonal influence on the bench press and squat explosive strength should be minimal. Finally, it must be noted that the findings of the present study may not be generalized to highly

strength-trained athletes, as this study included only female participants who engaged in recreational strength training.

Practical Applications

We showed that HIIT cycling (4×4 min at 80% of MAP) impaired lower but not upper-body explosive strength. Hence, practitioners should be cautious when programming HIIT and explosive strength exercise for the same muscle group. Since HIIT does not attenuate upper-body explosive strength, lower-body HIIT and upper-body explosive strength can be trained within close proximity without compromising the training quality. However, combining both HIIT cycling and maximal squat exercise (eg, $3 \times 5 + 3 \times 3$ at 80% of 1RM) appears to impair the subsequent explosive bench performance. Moreover, when interpreting these findings, it needs to be considered that the observed effects are specific to the HIIT (ie, 25 min) and strength protocol used, as well as to the population included. Practitioners should, therefore, be aware that the acute effects of combined HIIT and strength training shown in this study may not be generalizable to highly strength-trained or even competitive female athletes, and further research considering other female populations is warranted.

Conclusions

Our data indicate that lower- but not upper-body explosive strength is reduced by lower-body HIIT in naturally menstruating women who are recreationally strength trained. However, lower-body HIIT combined with upper- or lower-body strength loading resulted in a reduction of both squat and bench-press explosive strength. This finding indicates that, at least to some extent, central fatigue mechanisms may induce reductions in explosive strength performance of unloaded muscle groups following combined endurance and strength training sessions in naturally menstruating women. Importantly, the magnitude of reductions in explosive strength in the 2 loadings was similar to previous findings shown in men. Since the magnitude of fatigue might be dependent on the menstrual cycle phase and only the low hormone phase was tested in this study, future research should assess whether these findings are similar for different hormonal environments (eg, luteal phase). Furthermore, longitudinal studies are needed in order to assess whether our findings can be transferred to chronic training adaptations.

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